AMMAR ELTIGANI

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EDUCATION

University of Michigan
Marjorie Lee Browne Scholar
M.S., Mathematics — *August 2024 - May 2026*
Ph.D., Mathematics — *May 2026 - Present*

Williams College — *September 2019 - January 2024*
B.A. Mathematics, Computer Science — Overall GPA: 3.87/4

University of Oxford — *September 2021 - June 2022*
Williams-Exeter Programme at Oxford Fellow

EXPERIENCE

Math 115/116 (Calculus I/II) Graduate Student Instructor, University of Michigan MI
August 2024 - Present

- Primary instructor for a section of inductory calculus . Responsibilities include: organizing and delivering lectures 3x/week, writing and grading quizzes, grading exams, and holding office hours 3x/week.

MaCSS Summer Institute Graduate Student Assistant, University of Michigan MI June 2025 - August 2025

- Taught basic probability crash course.
- Mentored two undergraduate teams working on data science projects.
- Facilitated weekly group study sessions for MaCSS scholars.

SMALL Research Experience for Undergraduates (REU), Williams College MA
June 2023 - August 2023
Research in Mathematics with Professor Susan Loepp

- Worked in commutative algebra on completions of local rings.
- Characterized completions of Quasi-Excellent domain (link, *Contributions to Algebra and Geometry*).
- Extended a result about formal fibers (link, *Journal of Algebra and Its Applications*).
- Presented a poster (link) and gave an invited talk at the Young Mathematician's Conference (link).

Google Quantum AI, Venice CA — June 2022 - September 2022
Research Software Engineering Internship

- Added a major feature to Google's open-source quantum circuit compiler Cirq (link).
- Designed (link) and implemented (link) a qubit routing algorithm based on arxiv:1902.08091.
- Improved performance and runtime of existing implementation by 100x, outperforming IBM and t|ket>.

Google, Remote — June 2021 - September 2021
STEP Internship

- Enhanced a suite of performance testing products with RPC Record/Replay functionality
- Worked in Google's Java stack (Java App Dev Frameworks, Guice), other internal Google tools
- Successfully the ambiguity, flushing out design details and alternative solutions that arose from our project needing to integrate with many other existing testing products/teams

AI Researcher, Oxford AI Society, UK September 2021 - December 2021
Research in Machine Learning

- Selected as an Oxford AI Society undergraduate research fellow
- Explored artistic applications of large generative machine learning models
- Built from first principles an "AI Band" that can generate music, lyrics, and video for songs using Python and Blender

Research Assistant, Remote June 2020 - Aug 2020
Research in Algorithm Design

- Worked on Edit Distance Similarity Search using Locality Sensitive Hashing and Suffix trees.
- Tested and designed different data sets on state-of-the-art Similarity algorithms.
- Implemented suffix tree solution that improves on a trade-off between accuracy vs precision.

Teaching Assistant, Williams College MA September 2020 - December 2023

- Led help sessions and graded weekly student homework.
- Math courses TA-ed: Applied Real Analysis; Abstract Algebra.
- CS courses TA-ed: Data Structures & Advanced Programming; Algorithm Design & Analysis.

UNDERGRADUATE THESIS

The Khovanskii polynomial in $\mathbb{Z}^d$ February 2023 - Present
Advisor: Leo Goldmakher (defense, manuscript) *Williams College*

- Learned about various tools and ideas in: additive combinatorics, Ehrhart theory, residues on lattices, the geometry of numbers, and convex polytopes.
- Derived explicit binomial formula for the Khovanskii polynomial of a finite set $A \subset \mathbb{Z}^d$.
- Bounded the phase transition of the Khovanskii polynomial for special families of sets.

TALKS & CONFERENCES

UofM Summer School in Random Matrix Theory '26: [upcoming] (link)

Northwestern Summer School in high-dimensional probability '26: [upcoming] (link)

Marjorie Lee Browne Symposium Talk '26: *Random Matrix Theory for High-Dimensional Machine Learning (slides, paper)*

UofM Combinatorics Student Seminar '25: *Stability of the Size of Sumsets*

BRIDGES '24: Conference on graphical designs, algebraic statistics, and geometric group theory

JMM '24: *Controlling Formal Fibers of Countably Many Principal Prime Ideals* (slides)

JMM '24: *Completions of Quasi-Excellent Domains* (slides)

Young Mathematician's Conference '23: *Completions of Quasi-Excellent Domains* (slides)

Google Quantum AI '22: *Qubit Routing in Cirq* (slides)

CS Colloquium '20 (Williams College): *A suffix tree solution to the Similarity Search problem* (slides)

EXPOSITORY WRITING & OTHER RESEARCH

An empirial study of approximation algorithms for top-lists aggregation methods (link)

An introduction to perfect graphs (link)

A suffix tree solution to the similarity search/joins problem (lightning colloquium talk link)

An introduction to symmetric polynomials through basic invariant theory (link)

Some (old) projects on Github: https://github.com/ammareltigani

RELEVANT COURSES

Algorithm Design and Analysis
Intro to Probability
Machine Learning (MIT 6.036, online)
Computational Algebraic Geometry
Complex Analysis
Combinatorics and Graph Theory (graduate)
Differential Topology (graduate)
Topics in Convexity (graduate)
Probability & Stochastic Processes I (graduate)
Topics in Math of Data Science (graduate)
Probability & Stochastic Processes II (graduate)
Partial Differential Equations (graduate, current)
Data Structures & Advanced Programming
Quantum Computation & Information (graduate)
Quantum Software (graduate)
Algorithmic Game Theory
Analytic Number Theory
Algebra I (graduate)
Algebraic Topology (graduate)
Analysis II (Measure Theory) (graduate)
Convex Optimization (graduate)
Functional Analysis (graduate)
Machine Learning (graduate, current)
Reinforcement Learning (graduate, current)

TOOLS AND LANGUAGES

Natural Languages	English, Arabic, French, (West African) Krio
Computer Languages	Python (scientific/data stack)
Software	LaTeX, Mathematica, SageMath, Git/Bash

OTHER

Herchel-Smith Fellow (a highly competitive 2-6 year fully funded scholarship to pursue graduate studies at the University of Cambridge; awared by Williams College)

Underrepresented Identities in Computer Science (UNICS) tutor

Google Computer Science Research Mentorship Program (CSRMP) fellow